

Aran Mohammad

Modeling and Control Engineer

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Experience

03/2021 – 11/2025	Research Associate , Institute of Mechatronic Systems, Hannover - Led a project for development of real-time control for mechatronic systems (Link) - Implementation of algorithms (MATLAB/Simulink, C++) and automated code generation for real-time systems (Linux RT, Speedgoat with I/O modules) - Real-time integration of state observers (Kalman filters) for sensor fusion and machine-learning methods as safety functions - Physical modeling of kinematics/kinetics (Maple) including computationally optimal code generation and logic design using state machines (Stateflow) - Integration of sensor and drive technology via EtherCAT/EtherLab - Commissioning of 6 safety-critical test benches including SiL/HiL tests - Successful project acquisition (€400,000), presentation of results to industry partners, mentoring of 40 students/theses, and conducting courses in English
03/2020 – 11/2020	Research Intern and Master Student , IAV GmbH, Gifhorn Physics-based and data-driven emission prediction (Links 1 and 2)
10/2019 – 12/2019	DAAD-funded Research Intern , UNESP, Bauru, Brazil Frequency domain analysis and system identification of CFRP structures in MATLAB
10/2018 – 12/2018	Student Research Project , IAV GmbH, Gifhorn Design of a state observer with road grade estimation for ADAS in MATLAB/Simulink
10/2014 – 01/2018	Tutor for Control and Mechanics , Leibniz University Hannover (LUH)

Education

since 03/2021	Ph.D. candidate in Mechanical Engineering, LUH
10/2017 – 12/2020	M.Sc. in Mechanical Engineering (Grade: 1.0, with distinction), LUH
04/2014 – 09/2017	B.Sc. in Mechanical Engineering (Grade: 2.2), LUH

Awards

Master's Degree	M.Sc. Mechanical Engineering with distinction (GPA: 1.0)
Dean's List	2nd out of 238 students in M.Sc. Mechanical Engineering
Research Award	Best Paper Award for an outstanding scientific publication

Skills

Software	Excellent: MATLAB/Simulink, Python, ROS, Git, MS Office Very good: C++, C, SolidWorks, Autodesk Inventor, Maple, Linux Familiar with: PLC (TIA Portal, TwinCAT), IEC 61131-3, IBM Doors, SysML
Libraries	scikit-learn, PyTorch, Tensorflow, SciPy, MuJoCo, OpenCV
Languages	German (native), English (C1), French (B1), Kurdish (native)

Projects

System Design	System architecture for mechatronic systems - Implementation of a modular codebase for multiple test benches with EtherCAT - Integration of camera, force, and inertial sensors via Python, C++ and ROS - Publication: aranmoham.github.io/SafePR/
Research	Model-based control - Implementation of real-time capable algorithms in MATLAB/Simulink (PDF-Link) - Extended Kalman Filter for sensor fusion and state estimation (PDF-Link) - Design of redundant safety functions (logic, kinematics, dynamics) - 7 publications on control of mechatronic systems (Google Scholar Link)